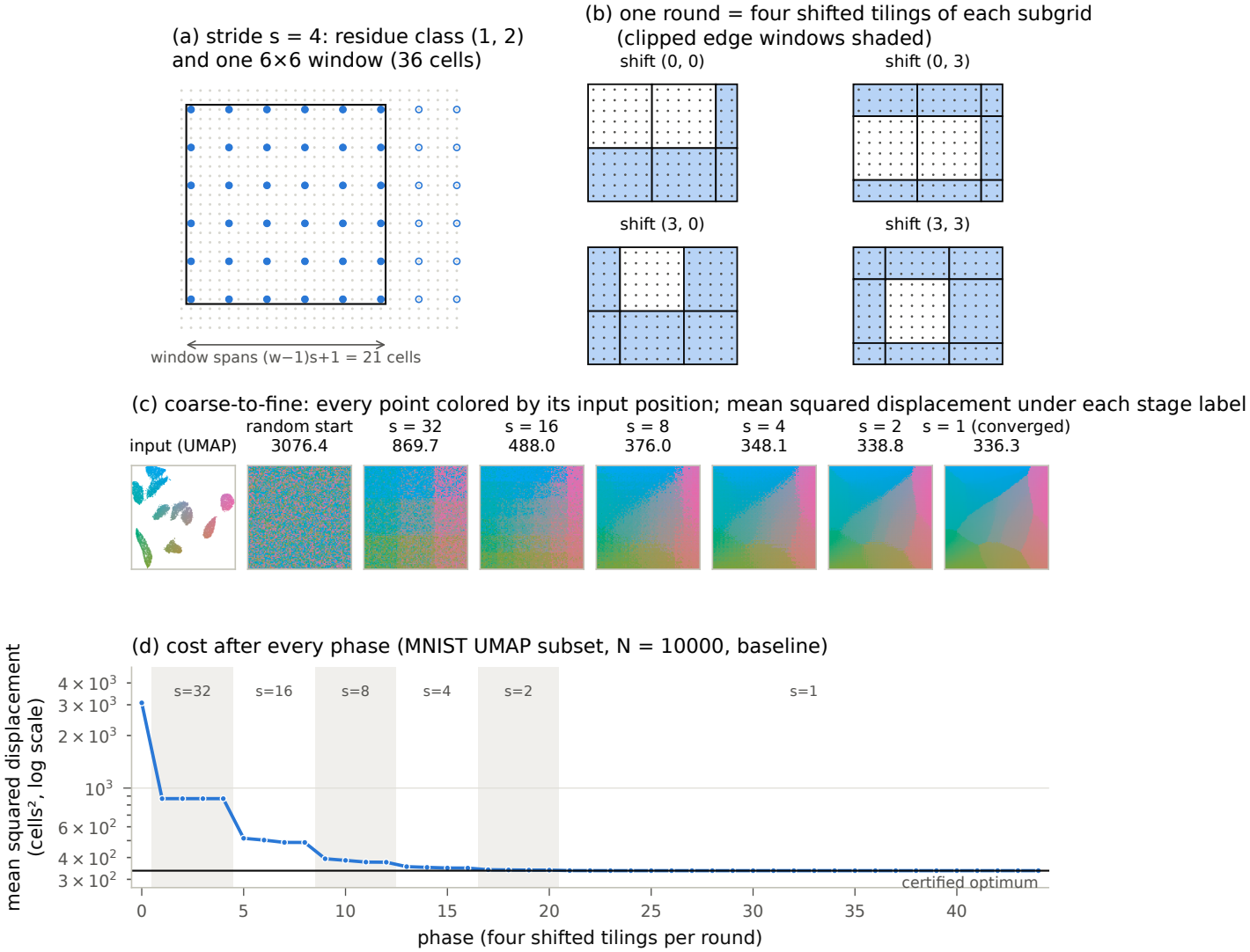

(a) stride s = 4: residue class (1, 2)
and one 6×6 window (36 cells)
window spans (w−1)s+1 = 21 cells
(b) one round = four shifted tilings of each subgrid
(clipped edge windows shaded)
shift (0, 0)
shift (0, 3)
shift (3, 0)
shift (3, 3)
(c) coarse-to-fine: every point colored by its input position; mean squared displacement under each stage label
input (UMAP)
random start
3076.4
s = 32
869.7
s = 16
488.0
s = 8
376.0
s = 4
348.1
s = 2
338.8
s = 1 (converged)
336.3
(d) cost after every phase (MNIST UMAP subset, N = 10000, baseline)
mean squared displacement
(cells², log scale)
4 × 10³
3 × 10³
2 × 10³
10³
6 × 10²
4 × 10²
3 × 10²
s=32
s=16
s=8
s=4
s=2
s=1
certified optimum
0
5
10
15
20
25
30
35
40
phase (four shifted tilings per round)